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Introduction to Quantum Computing

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In these lecture notes, a short introduction to quantum computing is provided for readers with a background in applied mathematics. The focus of these notes is to provide the tools to understand gate-based quantum computing. The advantage of quantum computing is illustrated on the *quantum Fourier transform* (QFT) which is a quantum equivalent to the discrete Fourier transform, implemented with a polylog number of elementary gates. As an application of the QFT, the *quantum phase estimation* (QPE) is presented. This quantum algorithm enables to compute an eigenvalue of a Hermitian matrix. If this Hermitian matrix (or more precisely, its unitary evolution) can be efficiently implemented on a quantum computer, the QPE offers an example of a quantum advantage.

These notes are largely inspired by the lecture notes of Lin Lin Lin [Lin22] and by the book of Nielsen and Chuang Nielsen and Chuang [NC10], which are recommended for a more complete treatment of the subject.

1 Tensor products

1.1 Tensor product of two vector spaces

$\|\cdot\|$ denotes the Euclidean norm on $\mathbb{C}^n$. * denotes the complex conjugate for numbers and the complex transpose for matrices.

Let $A \in \mathbb{C}^{m\times n}$ be a matrix. A defines a bilinear map from $\mathbb{C}^m \times \mathbb{C}^n$ to $\mathbb{C}$ via $(u, v) \mapsto \langle u, Av\rangle$. We can define a linear map that corresponds to this bilinear map. Let $(e_i), (f_j)$ be the canonical vectors of $\mathbb{C}^m$ and $\mathbb{C}^n$. Let $\mathcal{L}$ be the linear map defined for $1 \le i \le m, 1 \le j \le n$ by

$$\mathcal{L}(e_i \otimes f_j) = \langle e_i, Af_j\rangle,$$

and extended by linearity. The space $\mathrm{Span}(e_i \otimes f_j)$ is independent of the bilinear map A considered. Its dimension is thus the same as the space of bilinear maps on $\mathbb{C}^m \times \mathbb{C}^n$, which is equal to mn. The space $\mathbb{C}^m \otimes \mathbb{C}^n$ is thus defined as the span $\mathrm{Span}(e_i \otimes f_j)$.

Remark 1.1. A more informal way to view this construction is to view the tensor product $\mathbb{C}^m \otimes \mathbb{C}^n$ as the space of flattening of matrices. The definition of the linear map $\mathcal{L}$ is the flattening of the Frobenius scalar product on $\mathbb{C}^{m\times n}$, since $\langle e_i, Af_j\rangle = \mathrm{Tr}(Af_j e_i^*)$.

The tensor product of matrices of two matrices $A \in \mathbb{C}^{m\times n}, B \in \mathbb{C}^{p\times q}$ is defined likewise, on $\mathbb{C}^n \otimes \mathbb{C}^q$ by

$$(A \otimes B)(u \otimes v) = (Au) \otimes (Bv), \ \forall\, u, v \in \mathbb{C}^n \times \mathbb{C}^q$$

and extended by linearity. One can check that the matrix elements of this linear map is given by the Kronecker product formula.

Definition 1.1 (Tensor product of matrices)**.** For $A \in \mathbb{C}^{m\times n}$ and $B \in \mathbb{C}^{p\times q}$, the matrix $A \otimes B \in \mathbb{C}^{mp\times nq}$ has the following matrix elements:

$$(A \otimes B)_{ij,k\ell} = A_{ik}B_{j\ell}, \quad \text{for } 1 \le i \le m, 1 \le j \le p, 1 \le k \le n, 1 \le \ell \le q.$$

Proposition 1.1. *Let A, B, C, D be matrices of compatible sizes. The following assertions are true*

- *for $\lambda_A, \lambda_B, \mu_C, \mu_D \in \mathbb{C}$,*
$$(\lambda_A A + \lambda_B B) \otimes (\mu_C C + \mu_D D) = \lambda_A\mu_C A \otimes C + \lambda_A\mu_D A \otimes D + \lambda_B\mu_C B \otimes C + \lambda_B\mu_D B \otimes D$$
- $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$
- $(A \otimes B)^* = A^* \otimes B^*$
- *if A and B are unitary matrices, then $A \otimes B$ is a unitary matrix*
- *if (U_A, Σ_A, V_A^*) and (U_B, Σ_B, V_B^*) be singular value decompositions of A and B, then $(U_A \otimes U_B, \Sigma_A \otimes \Sigma_B, V_A^* \otimes V_B^*)$ is a singular value decomposition of $A \otimes B$.*

The first three statements are consequences of the definition of a Kronecker product. The last two statements directly follow from the second and third statements.

1.2 Extension to several vector spaces

The power of quantum computing lies in handling several tensor product of $\mathbb{C}^2$.

The tensor product of several vector spaces can be defined inductively as

$$\mathbb{C}^{n_1} \otimes \mathbb{C}^{n_2} \otimes \cdots \otimes \mathbb{C}^{n_d} = \left(\mathbb{C}^{n_1} \otimes \mathbb{C}^{n_2}\right) \otimes \cdots \otimes \mathbb{C}^{n_d}.$$

The tensor product of several matrices is defined likewise

$$(A_1 \otimes \cdots \otimes A_d)(u_1 \otimes \cdots \otimes u_d) = (A_1 u_1) \otimes \cdots \otimes (A_d u_d).$$

Notation: for $A \in \mathbb{C}^{m \times n}$ and $p \in \mathbb{N}$, we will write $A^{\otimes p} = \underbrace{A \otimes \cdots \otimes A}_{p \text{ times}}$.

2 Basics of Quantum Mechanics

2.1 Quantum State

Definition 2.1 (Quantum state)**.** Let $N \in \mathbb{N}$. Let $\sim$ be the equivalence relation on $\mathbb{C}^N$ defined by

$$\forall\, \phi, \psi \in \mathbb{C}^N, \phi \sim \psi \Leftrightarrow \exists \lambda \neq 0 \in \mathbb{C}, \psi = \lambda \phi.$$

Let $\mathrm{P}\mathbb{C}^N$ be the quotient set $\mathbb{C}^N$ by $\sim$. A quantum state is an element of $\mathrm{P}\mathbb{C}^N$.

Remark 2.1 (Dirac notation). For $\psi \in \mathbb{C}^N, \psi \neq 0$, the corresponding quantum state is denoted by $|\psi\rangle$. For $\phi \in (\mathbb{C}^N)^*, \phi \neq 0$, the dual quantum state is denoted by $\langle\phi|$. The scalar product between $|\phi\rangle$ and $|\psi\rangle$ is denoted by $\langle\phi|\psi\rangle$. If $\phi, \psi \in \mathbb{C}^N$ are such that $\|\phi\| = \|\psi\| = 1$ then $\langle\phi|\psi\rangle = \sum_{i=1}^N \phi_i^* \psi_i$.

In the following, we will identify $|\psi\rangle$ to a normalised representing vector $\psi \in \mathbb{C}^N$. Hence we will write the coordinates of $|\psi\rangle$ in $\mathbb{C}^N$ as $\begin{bmatrix} \psi_1 \\ \vdots \\ \psi_N \end{bmatrix}$, where $\sum_{i=1} |\psi_i|^2 = 1$. Note that by definition of the equivalence class, all vector representations of the form $\begin{bmatrix} e^{\mathrm{i}\theta}\psi_1 \\ \vdots \\ e^{\mathrm{i}\theta}\psi_N \end{bmatrix}$, for $\theta \in \mathbb{R}$, are equivalent.

Remark 2.2. For $|\phi\rangle, |\psi\rangle$ quantum states, $|\psi\rangle\langle\phi|$ denotes the projection onto ψ parallel to ϕ and is identified with a matrix $\mathbb{C}^{N \times N}$.

2.2 Evolution of a quantum state

For $|\psi\rangle, |\phi\rangle$ quantum states, there exists a matrix $U \in \mathbb{C}^{N \times N}$ such that $|\psi\rangle = U|\phi\rangle$.

Definition 2.2 (Schrödinger equation)**.** Let $(H(t))_{t \geq 0}$ be a family of Hermitian matrices, and $|\psi_0\rangle$ a quantum state. The Schrödinger evolution of a quantum state $t \mapsto \psi(t)$ is defined as the solution to the equation

$$\begin{cases} \mathrm{i}\dfrac{\partial}{\partial t}\psi(t) = H(t)\psi(t), \quad t \geq 0 \\ \psi(0) = \psi_0 \end{cases}$$

One can check that if $\|\psi_0\| = 1$ then for all $t \geq 0$, $\|\psi(t)\| = 1$.

If $t \mapsto H(t)$ is constant and equal to H, then the solution to the Schrödinger equation is $\psi(t) = e^{\mathrm{i}Ht}\psi_0$. This shows that for Hermitian matrices H, the matrix $e^{\mathrm{i}H}$ is a unitary matrix. Conversely, one can show that any unitary matrix U can be written as $e^{\mathrm{i}H}$ for some Hermitian matrix H.

2.3 Observables and measurements

Definition 2.3 (Observable)**.** An observable $O \in \mathbb{C}^{N\times N}$ is a Hermitian matrix. For $|\psi\rangle$ a quantum state, the expectation value denoted by $\langle\psi|O|\psi\rangle$ is equal to $\langle\psi, O\psi\rangle$.

Let O be an observable. Then O can be written in the basis of its eigenvectors as $O = \sum_{m=1}^{N} \lambda_m P_m$, where for $1 \leq m \leq N$, P_m orthogonal projector, $P_m = |u_m\rangle\langle u_m|$. A quantum state $|\psi\rangle$ can be written in the basis (u_m) as $\psi = \sum_{m=1}^{N} p_m u_m$, with $\sum_{m=1}^{N} |p_m|^2 = 1$. Then the expectation value of the observable is $\langle\psi|O|\psi\rangle = \mathrm{Tr}(|\psi\rangle\langle\psi|O) = \sum_{m=1}^{N} \lambda_m |p_m|^2$.

Definition 2.4 (Measurement)**.** Let $|\psi\rangle$ be a quantum state on $\mathbb{C}^N$ and $O \in \mathbb{C}^{N\times N}$ be an observable. Let (u_m) be an orthonormal basis of eigenvectors of O. Let $(p_m) \in \mathbb{C}^N$ such that $|\psi\rangle = \sum_{k=1}^{N} p_m u_m$ (with $\sum_{k=1}^{N} |p_m|^2 = 1$). The measurement operator $\mathcal{M}$ for the observable O is the operator such that

$$\mathcal{M}|\psi\rangle = u_m, \quad \text{with probability } |p_m|^2.$$

The measurement is, for a general quantum state $|\psi\rangle$ and a general observable O, a nonlinear operator. The output of the measurement is probabilistic and depends on the expansion of the quantum state ψ in the basis of eigenvectors of the observable O. The action of a measurement alters the quantum state. This has important consequences in practice for quantum computers as error corrections for quantum computers are much more delicate than in classical computing.

Measurements are the only way to have access to the information contained in the quantum state $|\psi\rangle$. For efficient quantum computations, it is thus important that the quantum state obtained after quantum computations has contributions in a few eigenvectors of the measured observable.

3 Quantum Computer

3.1 Qubit

In classical computing, the unit of information is a boolean $b \in \{0, 1\}$. In quantum computing, the corresponding unit of information is an element of $P\mathbb{C}^2$, called the *quantum bit* (or *qubit*).

Definition 3.1 (Quantum bit (qubit))**.** A quantum bit (or qubit) is a quantum state on $\mathbb{C}^2$.

The canonical basis associated to the space of quantum bits on $\mathbb{C}^2$ is denoted by $|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}$ and $|1\rangle = \begin{bmatrix}0\\1\end{bmatrix}$.

A quantum bit $|\psi\rangle \in \mathbb{C}^2$ satisfies $|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle$ with $|\psi_0|^2 + |\psi_1|^2 = 1$ (up to a global phase).

A qubit can be parametrised by two angles: factorising the complex phase $e^{i\gamma}$ of ψ_0, then accounting for the complex phase difference $e^{i\varphi}$ between ψ_0 and ψ_1, we get

$$|\psi\rangle = e^{i\gamma}(\cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle),$$

where $0 \leq \theta \leq \pi$, $0 \leq \varphi < 2\pi$ (recall that the global phase $e^{i\gamma}$ is not relevant).

This leads to the representation on the Bloch sphere (see Figure 1):

Every point in the Bloch sphere is a different quantum state. This illustrate at the level of one-qubit the encoding power of a unit of information in quantum computing compared to classical computing. In classical computing, the unit of information is a boolean, whereas in quantum computing, it encodes two continuous variables.

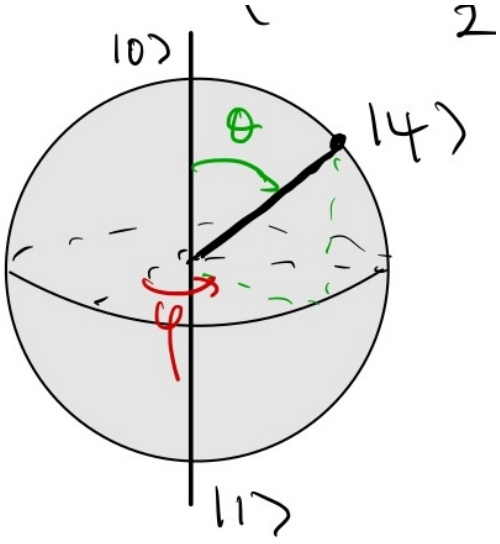


Figure 1: Bloch sphere

3.1.1 Pauli gates

Definition 3.2. Let $X, Y, Z \in \mathbb{C}^{2\times 2}$ be the Pauli matrices defined by

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Notice that the Pauli matrices X, Y, Z are Hermitian. They satisfy the commutation relations $[X, Y] = \mathrm{i}Z, [Y, Z] = \mathrm{i}X, [Z, X] = \mathrm{i}Y$. One can show that (I, X, Y, Z) is in fact a basis of the $\mathbb{R}$-vector space of Hermitian matrices. This means that unitary evolutions of quantum states can be written as $e^{\mathrm{i}\theta X + \mathrm{i}\phi Y + \mathrm{i}\gamma Z}$ (as the global phase can be ignored).

This motivates the introduction of parametrised gates

$$e^{\mathrm{i}\theta/2X} = \begin{bmatrix} \cos(\theta/2) & -\mathrm{i}\sin(\theta/2) \\ -\mathrm{i}\sin(\theta/2) & \cos(\theta/2) \end{bmatrix}, \quad e^{\mathrm{i}\theta/2Y} = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}, \quad e^{\mathrm{i}\theta/2(Z-I)} = \begin{bmatrix} e^{-\mathrm{i}\theta/2} & 0 \\ 0 & e^{\mathrm{i}\theta/2} \end{bmatrix}.$$

As the matrices X, Y, Z do **not** commute, we do **not** have $e^{\mathrm{i}\theta X + \mathrm{i}\phi Y + \mathrm{i}\gamma Z} = e^{\mathrm{i}\theta X} e^{\mathrm{i}\phi Y} e^{\mathrm{i}\gamma Z}$. However, using what is called *Trotter splitting*, we have that

$$e^{\mathrm{i}\theta X + \mathrm{i}\phi Y + \mathrm{i}\gamma Z} = \lim_{n\to\infty} \prod_{i=1}^{n} e^{\mathrm{i}\theta/nX} e^{\mathrm{i}\phi/nY} e^{\mathrm{i}\gamma/nZ},$$

which motivates the use of the parametrised gates introduced above.

3.1.2 Measurement of one qubit

Measurements of a qubit are taken for the observable Z (except otherwise stated). The computational basis $|0\rangle, |1\rangle$ is the basis that diagonalises Z, thus the measurement of a quantum state $|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle$, gives $|0\rangle$ with probability $p_0 = |\psi_0|^2$ and $|1\rangle$ with probability $p_1 = |\psi_1|^2 = 1 - p_0$.

This means that the measurement is a Bernoulli random variable with mean p_0 and variance $p_0(1 - p_0)$.

To estimate p_0, one uses the empirical estimator $\widehat{p}_0 = \frac{\text{number of measured } |0\rangle}{\text{total number of samples}}$. If $\mathcal{N}$ is the total number of samples, and assuming that each measurement is independent, the variance of the empirical estimator is $\frac{p_0(1-p_0)}{\mathcal{N}}$. To guarantee that the standard deviation of the empirical estimator is below some threshold ϵ, we must have $\frac{p_0(1-p_0)}{\mathcal{N}} \leq \epsilon^2$, thus $\mathcal{N} \geq \frac{p_0(1-p_0)}{\epsilon^2}$.

For a small probability p_0, an accurate estimation of the probability p_0 requires to have $\epsilon \ll p_0$, which means that the number of samples needs to be much larger than $\frac{1}{p_0}$. This is another important restriction in designing efficient algorithms in quantum computing.

Remark 3.1. The previous measurement only enables to estimate the modulus square of the amplitudes. It is therefore not enough to distinguish between $|\psi_+\rangle = \psi_0|0\rangle + \psi_1|1\rangle$ and, for example, $|\psi_-\rangle = \psi_0|0\rangle + \psi_1|1\rangle$. To differentiate both quantum states, a measurement after applying $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ can be used. The state obtained in both cases are

$$H|\psi_+\rangle = \frac{\psi_0+\psi_1}{\sqrt{2}}|0\rangle + \frac{\psi_0-\psi_1}{\sqrt{2}}|1\rangle, \quad \text{and} \quad H|\psi_-\rangle = \frac{\psi_0-\psi_1}{\sqrt{2}}|0\rangle + \frac{\psi_0+\psi_1}{\sqrt{2}}|1\rangle.$$

If $|\psi_0+\psi_1|^2 \neq |\psi_0-\psi_1|^2$, we will get $|0\rangle$ with different probabilities.

3.1.3 Other important one-qubit gates

Function	Description	Matrix Representation	Usage
H	Hadamard gate	$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$	`H(qubit)`
S	Phase gate (or S gate)	$\begin{bmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{bmatrix}$	`S(qubit)`
T	T gate	$\begin{bmatrix} 1 & 0 \\ 0 & e^{\mathrm{i}\pi/4} \end{bmatrix}$	`T(qubit)`

3.2 Collection of many qubits or the quantum register

A classical register is a collection of classical bits $b = (b_0, \dots, b_{n-1})$, $b_i \in \{0,1\}$, $i = 0, \dots, n-1$.

Definition 3.3 (Quantum register)**.** A quantum register on n qubits is a quantum state on $\bigotimes_{i=0}^{n-1} \mathbb{C}^2$.

The space $\bigotimes_{i=0}^{n-1} \mathbb{C}^2$ can be identified with $\mathbb{C}^{2^n}$. Its canonical basis, also called the *computational basis* is the set of the canonical basis of $\mathbb{C}^{2^n}$. These vectors are denoted by $|q_0 \dots q_{n-1}\rangle$, where $q_k \in \{0,1\}$ for $0 \leq k \leq n-1$ (*i.e.* the qubits are indexed from 0 to $n-1$). They correspond to the j-th canonical vector, $j \in \{0, \dots, 2^n-1\}$, where $j = \sum_{k=0}^{n-1} q_k 2^k$ (*i.e.* (q_k) is the binary decomposition of j). It is also customary to denote the quantum state $|j\rangle$.

Example 3.1 (Kronecker products of qubits)**.** If $|\psi_1\rangle \in \mathbb{C}^2$, $|\psi_2\rangle \in \mathbb{C}^2$, $|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$ is a 2-qubit state.

$$|0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Remark 3.2 (Entanglement)*.* The quantum states seen so far are given as Kronecker products. The power of quantum computing lies in the manipulation of *entangled* states, that **cannot** be obtained from Kronecker products of quantum states. For example, the *Bell state* $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)$ cannot be written as a Kronecker product of one qubit state.

Remark 3.3 (Superposition and entanglement). Entangled states are linear combinations of several Kronecker products of quantum states. Linear combinations of tensor product states are often called *superpositions*. A superposition of product states is not necessarily entangled. The state

$$H^{\otimes n}|0\rangle^{\otimes n} = (H|0\rangle) \otimes \cdots \otimes (H|0\rangle) = \frac{1}{2^{n/2}} \sum_{q_0,\dots,q_{n-1}\in\{0,1\}} |q_0 \dots q_{n-1}\rangle = \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} |k\rangle,$$

is a superposition of product states but is not entangled because it can be written as a product state (but in a different basis).

3.3 Gates

Linear operations on quantum registers are called *gates*. These linear operations are necessarily unitary operators.

In classical computing, any operation on classical registers can be implemented using combinations of only a few gates (for example, using only NAND or NOR gates).

In quantum computing, we would like to define an equivalent dictionary of elementary operations, whose compositions would generate any unitary operation. It is easy to check that *one-qubit gates*, defined as all the unitary operations of the form $I_2^{\otimes p} \otimes U \otimes I_2^{\otimes n-p-1}$ (with $0 \leq p \leq n-1$), are not enough to even approximate any unitary on $\mathbb{C}^{2^n}$. Indeed one can check that if one starts with $|0\rangle^{\otimes n}$, the application of one-qubit gates is equivalent to manipulate in parallel n one-qubit registers (by definition of the tensor product). It is necessary to add operations that act at least on two qubits, in order to be able to decompose any unitary on $\mathbb{C}^{2^n}$ in such elementary quantum operations.

The simplest two-qubit gate is the CNOT gate defined in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ by

$$\text{CNOT} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & & 1 \\ & & 1 & \end{bmatrix}$$

. The CNOT gate cannot be written as a tensor product of 2×2 unitaries. A simple way to check this assertion is to apply the CNOT gate on the following tensor product state

$$\text{CNOT}\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle\right) = \frac{1}{\sqrt{2}}\left(\text{CNOT}|00\rangle + \text{CNOT}|10\rangle\right) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

The outcome is the Bell state which cannot be written as tensor product.

It turns out that introducing all possible one-qubit gates and CNOT gates is sufficient to generate any unitary operation. Intuitively, this construction can be understood from the factorisation of any unitary matrix $U \in \mathbb{C}^{N\times N}$ as

$$U = G_1 \cdots G_k, D, \qquad k = \tfrac{N(N-1)}{2},$$

where $D = \text{diag}(e^{\mathrm{i}\theta_1}, \dots, e^{\mathrm{i}\theta_N})$ is a diagonal phase matrix and $(G_i)_{1\leq i\leq k}$ are Givens rotations (*i.e* identity except on a 2×2 block).

It is possible to further restrict this set, if we give up on the *exact* representation of the unitary U, *i.e.* for a given accuracy $\epsilon > 0$, we want to find $(U_1, \dots, U_m)$ such that

- $\|U - U_m U_{m-1} \cdots U_1\| < \epsilon$,
- $(U_k)_{1\leq k\leq m}$ belongs to a (small) set of universal gates.

There are multiple choices of universal gates such as $\{H, T, \text{CNOT}\}$ or $\{H, \text{Toffoli}\}$ (see their definitions below). A natural question is to check whether there are sets of universal gates that are better than others. The answer is given by the following theorem.

Theorem 3.1 (Solovay-Kitaev)**.** *Let $\mathcal{S}, \mathcal{T}$ be two sets of universal gates that are closed under inverses. Then any m-gate circuit using the gate set $\mathcal{S}$ can be implemented to precision ϵ using a circuit of $\mathcal{O}(m \operatorname{polylog}(m/\epsilon))$ gates from the gate set $\mathcal{T}$.*

Asymptotically, the Solovay-Kitaev theorem states that any choice of sets of universal gates leads to a comparable number of quantum operations.

3.4 Measurements for a quantum register

The measurement of a quantum register is defined as the measurement for the $Z^{\otimes n}$ operator, which is diagonal in the computational basis $|j\rangle_{0 \leq j \leq 2^n - 1}$.

Like in the one qubit case, for a quantum state $|\psi\rangle = \sum_{j=0}^{2^n-1} \psi_j |j\rangle$ with $\psi_j \in \mathbb{C}$, the probability of getting $|j\rangle$ after a measurement is equal to $|\psi_j|^2$.

In order to estimate $|\psi_j|^2$, many samples have to drawn from the discrete probabilities $(|\psi_k|^2)_{0 \leq k \leq 2^n - 1}$.

Remark 3.4 (Partial measurements). In the setting presented above, all the qubits are measured at the same time. It is possible to perform *partial measurements* on a subset of qubits. The wave-function formalism presented in these notes is not suited for a proper description of the outcome of partial measurements. It would require the introduction of the *density matrix* and the Heisenberg formulation of quantum mechanics, which encompasses the wave-function formalism.

3.5 No-copy Theorem

Theorem 3.2 (No copy)**.** *Let $|s\rangle \in \mathbb{C}^n$ be a quantum state. There is no unitary matrix $U \in \mathbb{C}^{n^2 \times n^2}$ such that for any $|\psi\rangle \in \mathbb{C}^n$, we have*

$$U|\psi\rangle \otimes |s\rangle = |\psi\rangle \otimes |\psi\rangle.$$

Proof. Let U be such that for any $|\psi\rangle \in \mathbb{C}^n$, we have

$$U|\psi\rangle \otimes |s\rangle = |\psi\rangle \otimes |\psi\rangle.$$

Let $|\phi\rangle \in \mathbb{C}^n$ such that $\langle\phi|\psi\rangle \neq 0$ or 1. Then $U|\phi\rangle \otimes |s\rangle = |\phi\rangle \otimes |\phi\rangle$. Thus we have

$$\begin{aligned} \langle\phi| \otimes \langle\phi|\psi\rangle \otimes |\psi\rangle &= \langle s| \otimes \langle\phi|U^*U|s\rangle \otimes |\psi\rangle \\ \langle\phi|\psi\rangle^2 &= \langle\phi|\psi\rangle. \end{aligned}$$

This means that either $\langle\phi|\psi\rangle = 0$ or $\langle\phi|\psi\rangle = 1$: contradiction. □

This theorem has a crucial consequence in the design of quantum algorithms: a vast majority of classical algorithms are iterative (Newton, root finding, conjugate-gradient…) which requires to store a copy of some iterate. In quantum computing, as it is impossible to copy arbitrary quantum states, the design of quantum algorithms cannot be a simple transposition of efficient classical algorithms.

Following the proof of the no-cloning theorem, it is however possible to copy *some* quantum states:

- for known quantum states $|\psi\rangle$, $|s\rangle$, if U_ψ is a unitary such that $U_\psi|s\rangle = |\psi\rangle$, then $(I \otimes U_\psi)(|\psi\rangle \otimes |s\rangle) = |\psi\rangle \otimes |\psi\rangle$;
- for quantum state $|j\rangle$ in the computational basis, then any state $|k\rangle$ in the computational basis can be copied, *i.e.* for any $0 \leq j \leq n-1$, there is U such that for any $0 \leq k \leq n-1$, we have

$$U|k\rangle \otimes |j\rangle = |k\rangle \otimes |k\rangle.$$

For $n = 4$ and $j = 0$, this unitary is the CNOT matrix: $\text{CNOT}(|k\rangle \otimes |0\rangle) = |k\rangle \otimes |k\rangle$. The CNOT matrix is not a copy operator (as the theorem forbids it). One can check that if $|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle$ with $\psi_0\psi_1 \neq 0$, then $\text{CNOT}|\psi\rangle \otimes |0\rangle = \psi_0|00\rangle + \psi_1|11\rangle \neq |\psi\rangle \otimes |\psi\rangle$.

3.6 Common quantum gates, measurement with MyQLM

Here is a list of common quantum gates as well as the corresponding command in MyQLM. The package MyQLM is a Python package that emulate the behaviour of a quantum computer (sometimes also called quantum processing unit or QPU). A quantum algorithm is simply a series of quantum gates applied to a initial quantum state (which is generally the state $|0\rangle$).

3.6.1 Constant gates

Function	Description	Matrix Representation	Usage
X	Pauli-X gate, NOT gate	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	`X(qubit)`
Y	Pauli-Y gate	$\begin{bmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{bmatrix}$	`Y(qubit)`
Z	Pauli-Z gate	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	`Z(qubit)`
H	Hadamard gate	$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$	`H(qubit)`
S	Phase gate (or S gate)	$\begin{bmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{bmatrix}$	`S(qubit)`
T	T gate	$\begin{bmatrix} 1 & 0 \\ 0 & e^{\mathrm{i}\pi/4} \end{bmatrix}$	`T(qubit)`
CNOT	CNOT (Controlled NOT) gate	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	`CNOT(control_qubit, target_qubit)`
CCNOT	Toffoli gate (or CCNOT gate).	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$	`CCNOT(control_qubit1, control_qubit2, target_qubit)`
CSIGN	Controlled Sign or C-Z gate	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$	`CSIGN(control_qubit, target_qubit)`
SWAP	SWAP gate	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	`SWAP(qubit1, qubit2)`

Function	Description	Matrix Representation	Usage
SQRTSWAP	Square Root of SWAP gate. It creates a superposition of swapped and non-swapped states.	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2}(1+\mathrm{i}) & \frac{1}{2}(1-\mathrm{i}) & 0 \\ 0 & \frac{1}{2}(1-\mathrm{i}) & \frac{1}{2}(1+\mathrm{i}) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	`SQRTSWAP(qubit1, qubit2)`
ISWAP	iSWAP gate. It swaps the states of two qubits with a phase factor of i.	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & \mathrm{i} & 0 \\ 0 & \mathrm{i} & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	`ISWAP(qubit1, qubit2)`

In this list, the CNOT, CSIGN, SWAP, SQRTSWAP and ISWAP are unitary operators acting on two qubits. It can be proved that these unitary operations cannot be written as the Kronecker product of two one-qubit gates.

The Toffoli gate is a 3-qubit gate that also cannot be written as a Kronecker product of one-qubit gates.

3.6.2 Parametrised gates

Function	Description	Matrix Representation	Usage
RX(θ)	Rotation around the X-axis by an angle θ.	$\begin{bmatrix} \cos(\theta/2) & -\mathrm{i}\sin(\theta/2) \\ -\mathrm{i}\sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$	`RX(theta)(qubit)`
RY(θ)	Rotation around the Y-axis by an angle θ.	$\begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$	`RY(theta)(qubit)`
RZ(θ)	Rotation around the Z-axis by an angle θ.	$\begin{bmatrix} e^{-\mathrm{i}\theta/2} & 0 \\ 0 & e^{\mathrm{i}\theta/2} \end{bmatrix}$	`RZ(theta)(qubit)`
PH(φ)	Phase gate that leaves $\|0\rangle$ unchanged and maps $\|1\rangle$ to $e^{\mathrm{i}\varphi}\|1\rangle$.	$\begin{bmatrix} 1 & 0 \\ 0 & e^{\mathrm{i}\varphi} \end{bmatrix}$	`PH(phi)(qubit)`

3.7 Quantum circuit

Quantum circuits are graphical representations of quantum algorithms. They show how quantum gates are applied to qubits over time.

3.7.1 Qubits and Wires

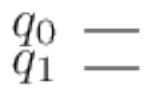
- **Qubits**: each qubit is represented by one horizontal line (a wire) in a circuit diagram.
- **Initial State**: qubits start in the $|0\rangle$ state unless otherwise specified.

Example in MyQLM

The following piece of code displays a quantum register initialised (by default) at $|00\rangle$.

```
from qat.lang.AQASM import Program
from qat.core.printer import plot_in_notebook
import matplotlib.pyplot as plt
```

```
my_program = Program()
qregister = my_program.qalloc(2) #allocates 2 qubits
circuit = my_program.to_circ()
plot_in_notebook(circuit,fmt='pdf') #circuit.display() works too
```



3.7.2 Gates

Gates are represented by boxes or dots in the quantum circuit. Their applications are read from left to right.

Example for one qubit in MyQLM

The following piece of code displays the quantum algorithm $XH|0\rangle$.

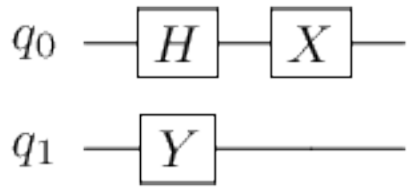
```
from qat.lang.AQASM import Program
from qat.lang.AQASM import H, X #import the Hadamard gate and the CNOT gate
my_program = Program()
qregister = my_program.qalloc(1) #allocates 2 qubits
H(qregister[0]) #first apply Hadamard gate
X(qregister[0]) #then apply X gate
circuit = my_program.to_circ()
plot_in_notebook(circuit,fmt='png') #circuit.display() works too
```



Example for two qubits in MyQLM

The following piece of code displays the quantum algorithm $(I \otimes Y)(X \otimes I)(H \otimes I)|00\rangle$.

```
from qat.lang.AQASM import Program
from qat.lang.AQASM import H, X, Y #import the Hadamard gate and the CNOT gate
my_program = Program()
qregister = my_program.qalloc(2) #allocates 2 qubits
H(qregister[0]) #first apply Hadamard gate on the first qubit
X(qregister[0]) #then apply X gate on the first qubit
Y(qregister[1]) #then apply X gate on the first qubit
circuit = my_program.to_circ()
plot_in_notebook(circuit,fmt='pdf') #circuit.display() works too
```

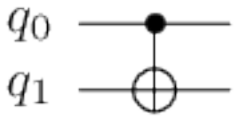


Note that since $(I \otimes Y)$ and $(X \otimes I)(H \otimes I)$, the quantum operations are squashed to the left.

CNOT gate in MyQLM

The following piece of code displays the quantum algorithm CNOT$|00\rangle$.

```
from qat.lang.AQASM import Program
from qat.lang.AQASM import CNOT #import the Hadamard gate and the CNOT gate
my_program = Program()
qregister = my_program.qalloc(2) #allocates 2 qubits
CNOT(qregister[0],qregister[1]) #first apply the CNOT gate with qubit 0 as the control and qubit 1 as the
circuit = my_program.to_circ()
plot_in_notebook(circuit,fmt='pdf') #circuit.display() works too
```



SWAP gate in MyQLM

The following piece of code displays the quantum algorithm SWAP$|00\rangle$.

```
from qat.lang.AQASM import Program
from qat.lang.AQASM import SWAP #import the Hadamard gate and the CNOT gate
my_program = Program()
qregister = my_program.qalloc(2) #allocates 2 qubits
SWAP(qregister[0],qregister[1]) #first apply the SWAP gate between qubit 0 and qubit 1
circuit = my_program.to_circ()
plot_in_notebook(circuit,fmt='pdf') #circuit.display() works too
```



3.7.3 Measurement

Statistical measurement of a quantum program can be done in MyQLM, by specifying the number of *shots*, i.e. the number of times that the program is executed and its output is measured. Each run collapses the quantum state $|\psi\rangle$ to a state of the computational basis and the observed frequencies approximate the probabilities $|c_{i_1 \ldots i_n}|^2$.

```
from qat.lang.AQASM import Program, H, CNOT
from qat.qpus import PyLinalg

prog = Program()
qreg = prog.qalloc(2)
prog.apply(H, qreg[0])
prog.apply(CNOT, qreg[0], qreg[1])
circ = prog.to_circ()

qpu = PyLinalg()
job = circ.to_job(nbshots=1000)  # 1000 measurements of the full register
result = qpu.submit(job)
for sample in result:
    print(sample.state, sample.probability)  # |00> and |11> with frequency ~1/2
```

It is also possible to retrieve the exact probabilities (and the exact amplitudes) of the encoded wave function, if needed. This feature offered by MyQLM cannot be reproduced with a true quantum computer.

```
job = circ.to_job()  # no nbshots: the exact wave function is returned
result = qpu.submit(job)
for sample in result:
    print(sample.state, sample.probability, sample.amplitude)  # |00> and |11> with probability exactly 1/
```

3.7.4 Complexity of a quantum algorithm

Remark 3.5 (Clifford gates). Clifford gates are quantum gates that stabilises the group formed by Kronecker products of Pauli matrices (*i.e.* if C is a Clifford gate and P a Kronecker product of Pauli matrices, then CPC^* is a Kronecker product of Pauli matrices). It can be checked that the Hadamard gate, the CNOT gate are Clifford gates, but not the T gate. Since these gates stabilise the group of Kronecker product of Pauli matrices, they can be classically simulated in polynomial time.

The complexity of quantum algorithms is estimated in three different ways:

- the depth of the circuit: the depth of the circuit is the maximal number of gates along any path from an input and an output. It is a reasonable depiction of the total run time of a quantum simulation. One of the main challenge in the design of physical quantum computers is in maintaining the *coherence time* of the quantum system. The coherence time is the longest period during which the quantum system accurately preserves the outcome of a quantum algorithm. Nowadays, the maximal coherence time is of the order of 0.1 ms and the total number of operations that can be performed is of the order $10^2, 10^3$.
- the number of two qubit gates in the quantum algorithm: CNOT gates have a probability of failure of about 1%, whereas one-qubit gates typically have failure rates of about 0.1% (or sometimes lower than that). The accuracy of a quantum algorithm depends heavily on the number of two-qubit gates.
- the number of T gates in the quantum algorithm: to a lesser extend, the complexity of a quantum algorithm can also be estimated by counting the number of T gates. If a quantum algorithm is expressed in the universal gate set $\{H, T, \text{CNOT}\}$, the only non-Clifford gate is the T gate. As Clifford gates are classically easy to emulate, the quantum advantage can be estimated as the total number of T gates.

4 Quantum Fourier Transform

The quantum Fourier transform is the quantum implementation of the discrete Fourier transform on n qubits.

4.1 Discrete Fourier Transform

We recall the definition of the discrete Fourier transform.

Definition 4.1 (Discrete Fourier transform)**.** For $N \in \mathbb{N}^*$, the Discrete Fourier Transform (DFT) F_N is defined by:

$$\forall 0 \leq j \leq N-1, \quad (F_N x)_j = \frac{1}{\sqrt{N}} \sum_{q=0}^{N-1} e^{2\mathrm{i}\pi \frac{qj}{N}} x_q \quad \text{where} \quad x = \begin{bmatrix} x_0 \\ \vdots \\ x_{N-1} \end{bmatrix}.$$

In the canonical basis, the DFT is represented by the matrix:

$$F_N = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \cdots & \omega^{(N-1)^2} \end{bmatrix}, \quad \text{where} \quad \omega = e^{2\mathrm{i}\pi/N}.$$

Proposition 4.1. *The matrix F_N satisfies $F_N^* F_N = I_N$, since:*

$$(F_N^* F_N)_{ij} = \frac{1}{N} \sum_{q=0}^{N-1} (\omega^*)^{iq} \omega^{jq} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Thus, F_N is an unitary matrix.

For $x \in \mathbb{C}^N$, $F_N x$ can be computed in $O(N \log N)$ using the *FFT algorithm* (faster than the naive $O(N^2)$ matrix-vector multiplication of a dense matrix).

4.2 Quantum Fourier transform

Definition 4.2 (Quantum Fourier Transform (QFT))**.** The quantum Fourier transform on n qubits is defined as the quantum gate such that for a quantum state $|\psi\rangle = \sum_{j=0}^{2^n-1} c_j |j\rangle$:

$$\mathrm{QFT}_n |\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} \left(\sum_{j=0}^{2^n-1} c_j e^{2\pi \mathrm{i} jk/2^n} \right) |k\rangle,$$

where $(|j\rangle)_{0 \leq j \leq 2^n-1}$ is the computational basis of $\mathbb{C}^{2^n}$.

The QFT is exactly the implementation in the computational basis of an n-qubit quantum computer of the discrete Fourier transform.

For a quantum state $|j\rangle$ of the computational basis, we have

$$\mathrm{QFT}_n |j\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi \mathrm{i} jk/2^n} |k\rangle.$$

Example 4.1. For one qubit, the QFT is the matrix $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. This coincides with the Hadamard matrix.

4.3 Implementation

4.3.1 Controlled rotation

Definition 4.3 (Controlled rotation)**.** Let $U \in \mathbb{C}^{2^n \times 2^n}$ be a unitary matrix. The controlled rotation for the unitary U, denoted C-U $\in \mathbb{C}^{2^{n+1} \times 2^{n+1}}$ is the matrix such that

$$\text{C-U} = |0\rangle\langle 0| \otimes I_{2^n} + |1\rangle\langle 1| \otimes U.$$

The qubit of index 0 is called the *control* qubit and the other qubits are called the *target* qubits.

One can check that the matrix C-U is a unitary matrix, and as such defines a quantum gate. One way to see that is to write the matrix in the computational basis

$$\text{C-U} = \begin{bmatrix} I_{2^n} & \\ & U \end{bmatrix},$$

where it is clear that C-U is unitary.

Remark 4.1. The CNOT gate is a controlled X gate.

Another way to see the controlled rotation is when it is applied to $|q\rangle \otimes |\psi\rangle$, where $|q\rangle$ is either $|0\rangle$ or $|1\rangle$. Then

$$\text{C-U}|q\rangle \otimes |\psi\rangle = |q\rangle \otimes U^q |\psi\rangle.$$

Proposition 4.2. *Let* $U \in \mathbb{C}^{2^n \times 2^n}, V \in \mathbb{C}^{2^m \times 2^m}$ *be unitary matrices. The controlled rotation* C-U $\otimes$ V $\in \mathbb{C}^{2^{n+m+1} \times 2^{n+m+1}}$ *is the matrix product of the controlled rotations*

$$\text{C-}(\text{U} \otimes \text{V}) = \text{C-}(\text{I}_{2^n} \otimes \text{V})\text{C-}(\text{U} \otimes \text{I}_{2^m}) = \text{C-}(\text{U} \otimes \text{I}_{2^m})\text{C-}(\text{I}_{2^n} \otimes \text{V}).$$

Proof. The equality is clear from the block decomposition of

$$\text{C-}(\text{U} \otimes \text{V}) = \begin{bmatrix} I_{2^{n+m}} & \\ & \text{U} \otimes \text{V} \end{bmatrix} = \begin{bmatrix} I_{2^{n+m}} & \\ & \text{I}_{2^n} \otimes \text{V} \end{bmatrix} \begin{bmatrix} I_{2^{n+m}} & \\ & \text{U} \otimes \text{I}_{2^m} \end{bmatrix}.$$

□

Qubit 0 is the control in the definition of the controlled rotation. This is arbitrary and controlled rotation can be defined for other qubits by sandwiching the controlled rotation with appropriate SWAPs.

Definition 4.4. Let $U \in \mathbb{C}^{2^n \times 2^n}$ be a unitary matrix. The controlled rotation with control qubit j for the unitary U, denoted $\text{C}_j\text{-U} \in \mathbb{C}^{2^{n+1} \times 2^{n+1}}$ is the matrix such that

$$\text{C}_j\text{-U} = \text{SWAP}_{0j}(\text{C-U})\text{SWAP}_{0j},$$

where SWAP_{0j} is the SWAP gate between qubits 0 and j defined by

$$\text{SWAP}_{0j}(|q_0 \dots q_{j-1} q_j q_{j+1} \dots q_{n-1}\rangle) = |q_j q_1 \dots q_{j-1} q_0 q_{j+1} \dots q_{n-1}\rangle, \quad \text{for any } q_0, \dots, q_{n-1} = 0, 1.$$

Suppose that U is a one-qubit gate, in general C-U $\neq$ C_1-U. An important case where it holds is the parametrised Z-rotation.

Proposition 4.3. *Let* $R_Z(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{\text{i}\theta} \end{bmatrix}$. *Then* C-$R_Z(\theta) = \text{C}_1$-$R_Z(\theta)$.

Proof. We have $\text{SWAP}_{01} = \begin{bmatrix} 1 & & & \\ & & 1 & \\ & 1 & & \\ & & & 1 \end{bmatrix}$ (in the $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ basis). In the same basis, we have C-$R_Z(\theta) = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & e^{\text{i}\theta} \end{bmatrix}$, thus the SWAP_{01} gate thus not affect the C-$R_Z(\theta)$ matrix. □

Definition 4.5 (Multi-controlled rotation)**.** Let $U \in \mathbb{C}^{2^n \times 2^n}$ be a unitary matrix. The multi-controlled rotation over m qubits, denoted $\text{C}^\text{m}\text{-U} \in \mathbb{C}^{2^{n+m} \times 2^{n+m}}$ is the matrix such that

$$\text{C}^\text{m}\text{-U} = \sum_{j=0}^{2^m-2} |j\rangle\langle j| \otimes I_{2^n} + |2^m - 1\rangle\langle 2^m - 1| \otimes U.$$

In the computational basis, a multi-controlled gate is written

$$\text{C}^\text{m}\text{-U} = \begin{bmatrix} I_{2^{m+n}-2^n} & \\ & U \end{bmatrix}$$

Remark 4.2. The Toffoli gate is a multi-controlled X gate over 2 qubits.

4.3.2 QFT circuit

Let $0 \leq j, k \leq 2^{n+1} - 1$. Let $(j_n, \dots, j_0)$ and $(k_n, \dots, k_0)$ be the binary representations of j, k (*i.e.* $j = \sum_{\ell=0}^{n} j_\ell 2^\ell$ and $k = \sum_{\ell=0}^{n} k_\ell 2^\ell$). Let $\overline{j_0} = \sum_{\ell=1} 2^{\ell-1} j_\ell$ and $\overline{k_n} = \sum_{\ell=0}^{n-1} 2^\ell k_\ell$.

We have

$$jk = (j_0 + 2\overline{j_0})(\overline{k_n} + 2^n k_n) = j_0 2^n k_n + j_0 \overline{k_n} + 2^{n+1} \overline{j_0} k_n + 2\overline{j_0 k_n}.$$

We compute the (j, k) entry of the discrete Fourier transform:

$$\begin{aligned}\exp(\frac{2\mathrm{i}\pi jk}{2^{n+1}}) &= \exp(\mathrm{i}\pi j_0 k_n) \exp(\frac{2\mathrm{i}\pi j_0 \overline{k_n}}{2^{n+1}}) \exp(2\mathrm{i}\pi \overline{j_0} k_n) \exp(\frac{2\mathrm{i}\pi \overline{j_0 k_n}}{2^n}) \\ &= (-1)^{j_0 k_n} \exp(\frac{2\mathrm{i}\pi j_0 \overline{k_n}}{2^{n+1}}) \exp(\frac{2\mathrm{i}\pi \overline{j_0 k_n}}{2^n}),\end{aligned}$$

where we have used that $j_0, \overline{j_0}, k_n, \overline{k_n}$ are integers.

We realise that the last coefficient is – up to a normalisation constant – the $(\overline{j_0 k_n})$ coefficient of the discrete Fourier transform on $\mathbb{C}^{2^n}$. This motivates the introduction of the matrix $\tilde{F}_{2^n} = F_{2^n} S$, where S is the permutation matrix of the flip of the indices, *i.e.* $S|q_0 \dots q_{n-1}\rangle = |q_{n-1} \dots q_0\rangle$ with $q_0, \dots, q_{n-1} = 0$ or 1. This matrix can be implemented using $\lfloor n/2 \rfloor$ SWAP gates.

If $j_0 = 0$, for $k_n = 0$ or 1, we see that $(\tilde{F}_{2^{n+1}})_{0\overline{j_0};k_n\overline{k_n}} = \frac{1}{\sqrt{2}} (\tilde{F}_{2^n})_{\overline{j_0};\overline{k_n}}$.

Let $P \in \mathbb{C}^{2^n \times 2^n}$ be the diagonal and unitary matrix such that $P_{kk} = \exp(\frac{2\mathrm{i}\pi k}{2^{n+1}})$. Then for $j_0 = 1$, we have that $(\tilde{F}_{2^{n+1}})_{1\overline{j_0};k_n\overline{k_n}} = \frac{1}{\sqrt{2}} (-1)^{k_n} (\tilde{F}_{2^n} P)_{\overline{j_0};\overline{k_n}}$.

The matrix $\tilde{F}_{2^{n+1}}$ has the following block structure in the $(|j_0\rangle|\overline{j_0}\rangle; |k_n\rangle|\overline{k_n}\rangle)$ basis

$$\tilde{F}_{2^{n+1}} = \frac{1}{\sqrt{2}} \begin{bmatrix} \tilde{F}_{2^n} & \tilde{F}_{2^n} \\ \tilde{F}_{2^n} P & -\tilde{F}_{2^n} P \end{bmatrix}.$$

This block structure can be factorised in

$$\frac{1}{\sqrt{2}} \begin{bmatrix} \tilde{F}_{2^n} & \tilde{F}_{2^n} \\ \tilde{F}_{2^n} P & -\tilde{F}_{2^n} P \end{bmatrix} = \begin{bmatrix} \tilde{F}_{2^n} & \\ & \tilde{F}_{2^n} \end{bmatrix} \begin{bmatrix} I_{2^n} & \\ & P \end{bmatrix} \begin{bmatrix} I_{2^n}/\sqrt{2} & I_{2^n}/\sqrt{2} \\ I_{2^n}/\sqrt{2} & -I_{2^n}/\sqrt{2} \end{bmatrix}.$$

The first matrix is simply $I_2 \otimes \tilde{F}_{2^n}$, the second matrix is a controlled-P gate and the last one is simply $H \otimes I_{2^n}$.

It suffices to determine the circuit for the rotation P in order to find the circuit of the QFT. As $P_{kk} = \exp(\frac{2\mathrm{i}\pi k}{2^{n+1}})$, using the binary writing $(k_0, \dots, k_{n-1})$ of the integer k, we have that

$$P_{kk} = \prod_{\ell=0}^{n-1} \exp(\frac{\mathrm{i}\pi k_\ell}{2^{n-\ell}}).$$

It is thus a Kronecker product of one qubit parametrised Z-rotations. Using Proposition 4.2, we deduce that the circuit can be written as

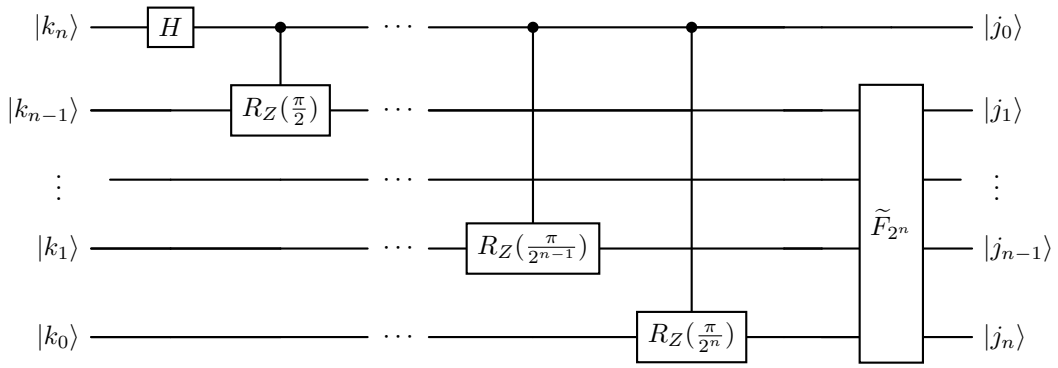


Figure 2: QFT circuit: first step

As the controlled rotations are parametrised-Z rotations, we have another (more common) writing of the algorithm as

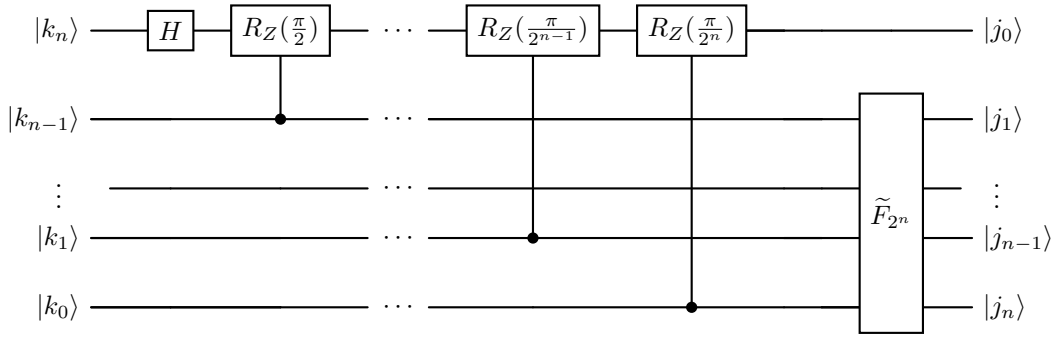


Figure 3: QFT circuit: first step

Performing the induction steps yields the full QFT circuit (not forgetting the SWAPs at the end):

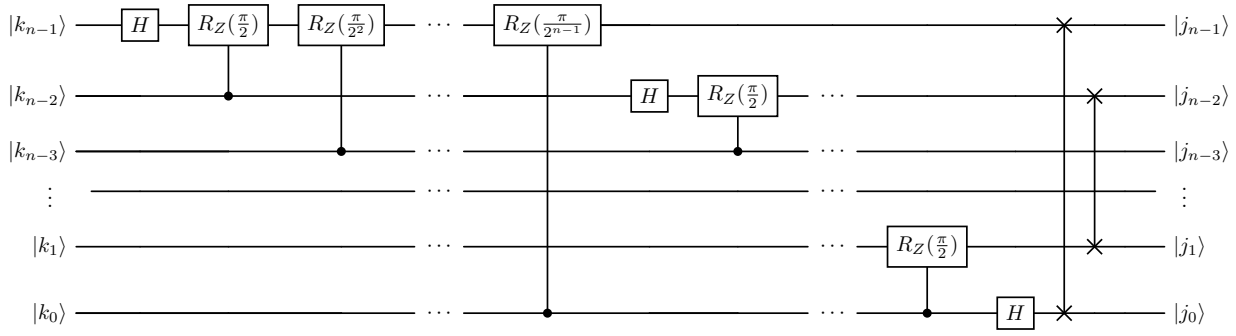


Figure 4: Full QFT circuit

Complexity: the QFT circuit has $\sum_{k=1}^{n-1} k = \frac{n(n-1)}{2}$ controlled gates and $\frac{n}{2}$ SWAPS, hence the complexity of applying a QFT is of the order n^2. This means that the discrete Fourier transform on $\mathbb{C}^{2^n}$ is exponentially faster (in terms of elementary operations) on a quantum computer, compared to its classical version.

5 Quantum Phase Estimation

Assumptions: the unitary matrix $U \in \mathbb{C}^{2^n \times 2^n}$ has an eigenvalue $e^{2\mathrm{i}\pi\theta}$ for a normalised eigenvector ψ such that there is $m \in \mathbb{N}$ such that $\theta = \frac{k}{2^m}$, with $0 \leq k \leq 2^m - 1$.

The goal of the quantum phase estimation (QPE) algorithm is to compute the eigenvalue $e^{2\mathrm{i}\pi\theta}$.

A typical example of a unitary U for which the eigenvalue is sought are operators of the form $e^{\mathrm{i}Ht}$ for a Hermitian matrix H.

Let $\mathcal{U} \in \mathbb{C}^{2^{n+m} \times 2^{n+m}}$ be the matrix defined by

$$\mathcal{U} = \sum_{j=0}^{2^m-1} |j\rangle\langle j| \otimes U^j.$$

A quick calculation shows that $\mathcal{U}$ is a unitary matrix.

The implementation of the matrix $\mathcal{U}$ can be simplified by realising that

$$\begin{aligned}
\mathcal{U} &= \sum_{j=0}^{2^m-1} |j\rangle\langle j| \otimes U^j \\
&= \sum_{j_0,\ldots,j_{m-1}} |j_0 \cdots j_{m-1}\rangle\langle j_0 \cdots j_{m-1}| \otimes U^{\sum_{\ell=0}^{m-1} j_\ell 2^\ell} \\
&= \sum_{j_0=0,1} |j_0\rangle\langle j_0| \otimes \cdots \otimes \sum_{j_{m-1}=0,1} |j_{m-1}\rangle\langle j_{m-1}| \otimes \prod_{\ell=0}^{m-1} U^{j_\ell 2^\ell}.
\end{aligned}$$

Definition 5.1 (Strong Kronecker product)**.** For $A \in \mathbb{C}^{mp\times mp}$, $B \in \mathbb{C}^{np\times np}$, we define the *strong Kronecker product* as the matrix denoted by $A \bowtie B \in \mathbb{C}^{mnp\times mnp}$ with

$$(A \bowtie B)_{abc,def} = \sum_{g=1}^{p} A_{ac;dg} B_{bc;gf}.$$

The strong Kronecker product is a mix of a Kronecker product with respect to the first index and a matrix-vector product in the second index.

Using this notation, we see that the expression of $\mathcal{U}$ can be written

$$\begin{aligned}
\mathcal{U} &= (\sum_{j_0=0,1} |j_0\rangle\langle j_0| \otimes U^{j_0}) \bowtie \cdots \bowtie (\sum_{j_{m-1}=0,1} |j_{m-1}\rangle\langle j_{m-1}| \otimes U^{2^{m-1} j_{m-1}}) \\
&= (\text{C-}U) \bowtie (\text{C-}U^2) \bowtie \cdots \bowtie (\text{C-}U^{2^{m-1}}).
\end{aligned}$$

The corresponding circuit is then given by

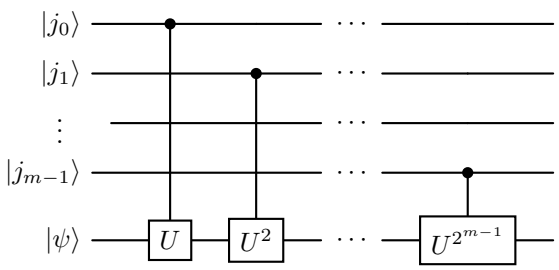


Figure 5: Circuit for $\mathcal{U}$

5.1 Circuit

Let $|\psi\rangle$ be an eigenvector of U for the eigenvalue $e^{2\mathrm{i}\pi\theta}$. Then for any $|\phi\rangle = \sum_{j=0}^{2^m-1} \phi_j |j\rangle$ (with $\sum_{j=0}^{2^m-1} |\phi_j|^2 = 1$), we have

$$\begin{aligned}\mathcal{U}|\phi\rangle \otimes |\psi\rangle &= \sum_{j=0}^{2^m-1} \phi_j |j\rangle \otimes U^j |\psi\rangle \\ &= \sum_{j=0}^{2^m-1} e^{2\mathrm{i}\pi\theta j} \phi_j |j\rangle \otimes |\psi\rangle.\end{aligned}$$

The coefficients of the quantum state in the first m qubits are given by the vector $(e^{2\mathrm{i}\pi\theta j}\phi_j)$. It can be viewed as the discretisation of a function $x \mapsto e^{2^{m+1}\mathrm{i}\pi\theta x}\phi(x)$ on the grid $x_j = \frac{j}{2^m}, j = 0, \dots, 2^m - 1$ (with $\phi(x_j) = \phi_j$).

As we are looking for the value θ, it is natural to consider the Fourier transform as if $x \mapsto e^{2^{m+1}\mathrm{i}\pi\theta x}$ is 1-periodic, then its inverse discrete Fourier transform is the canonical vector e_k where $1/k$ is the periodicity of the function $x \mapsto e^{2^{m+1}\mathrm{i}\pi\theta x}$.

The QPE circuit is given below

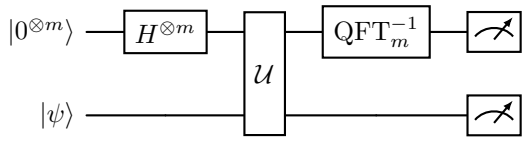


Figure 6: QPE circuit

Remark 5.1. For $m = 1$, the QPE circuit reduces to the Hadamard test $(H \otimes I_{2^n})(\text{C-}U)(H \otimes I_{2^n})$, which has the following circuit:

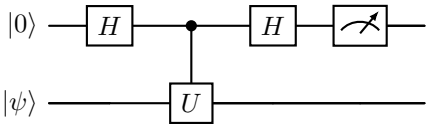


Figure 7: Hadamard test

As $H^{\otimes m}|0\rangle^{\otimes m} = \mathrm{QFT}_m|0\rangle^{\otimes m}$, the circuit can also be written with a QFT gate instead of the tensor product of Hadamard gates.

Theorem 5.1 (Quantum phase estimation)**.** *Assume that ψ is an eigenvector of the unitary U with eigenvalue $e^{2\mathrm{i}\pi\theta}$ such that $\theta = \frac{k}{2^m}$ for some $0 \leq k \leq 2^m - 1$.*

Then the resulting state of the QPE algorithm in Figure 6 is $|k\rangle \otimes |\psi\rangle$, i.e. *the result of the QPE algorithm is deterministic and only requires one measurement.*

Proof. By assumption on the eigenvalue, the coefficients of the quantum state in the first m qubits are the inverse discrete Fourier transform of the 1-periodic function $x \mapsto e^{2^{m+1}\mathrm{i}\pi\theta x}$. Its inverse discrete Fourier transform is thus the k-th canonical vector e_k.

We can also compute exactly the output of the circuit and find the same conclusion. By definition of the QFT, we have $\mathrm{QFT}_m|0\rangle^{\otimes m} = \frac{1}{2^{m/2}}\sum_{j=0}^{2^m-1}|j\rangle$. Thus we have

$$\begin{aligned}\mathcal{U}(\mathrm{QFT}_m|0\rangle^{\otimes m} \otimes |\psi\rangle) &= \frac{1}{2^{m/2}}\sum_{j=0}^{2^m-1}|j\rangle \otimes U^j|\psi\rangle \\ &= \frac{1}{2^{m/2}}\sum_{j=0}^{2^m-1} e^{2\mathrm{i}\pi\frac{kj}{2^m}}|j\rangle \otimes |\psi\rangle.\end{aligned}$$

Thus applying the last QFT_m^{-1} gate, we have

$$(\mathrm{QFT}_m^{-1} \otimes I_{2^n})(\frac{1}{2^{m/2}}\sum_{j=0}^{2^m-1} e^{2\mathrm{i}\pi\frac{kj}{2^m}}|j\rangle \otimes |\psi\rangle) = \frac{1}{2^m}\sum_{\ell=0}^{2^m-1}\sum_{j=0}^{2^m-1} e^{2\mathrm{i}\pi\frac{(k-\ell)j}{2^m}}|\ell\rangle^{\otimes m} \otimes |\psi\rangle.$$

Now $\sum_{j=0}^{2^m-1} e^{2\mathrm{i}\pi\frac{(k-\ell)j}{2^m}} = 0$ if $\ell \neq k$, thus the resulting state of the QPE is $|k\rangle \otimes |\psi\rangle$. □

5.2 Limitations and sources of error

There are two main limitation in the algorithm:

- an exact eigenvector ψ of the unitary matrix U needs to be implemented in the quantum computer: in practice, it is unreasonable to assume that such an eigenvector can be achieved exactly as a starting point for the QPE. One can only hope for a starting quantum state $|\tilde{\psi}\rangle$ which has a significant overlap with the target quantum state $|\psi\rangle$. In practice, it can be a challenge to even prepare such a state.
- another limitation in the algorithm is the m-digit representability of the angle θ associated to the eigenvalue $e^{2\mathrm{i}\pi\theta}$.

Let us investigate this point. As being stated above, the coefficients are the inverse discrete Fourier transform of the function $x \mapsto e^{2^{m+1}\mathrm{i}\pi x}$ on the grid points $(\frac{j}{2^m})_{0\leq j\leq 2^m-1}$. The modulus of the k-th coefficient is:

$$\begin{aligned}\Big|\frac{1}{2^{m/2}}\sum_{\ell=0}^{2^m-1} e^{2\mathrm{i}\pi\theta\ell}e^{-2\mathrm{i}\pi\frac{k\ell}{2^m}}\Big| &= \frac{1}{2^{m/2}}\Big|\sum_{\ell=0}^{2^m-1} e^{2\mathrm{i}\pi\frac{(2^m\theta-k)\ell}{2^m}}\Big| \\ &= \frac{1}{2^{m/2}}\Big|\frac{1-e^{2^{m+1}\mathrm{i}\pi\theta}}{1-e^{2\mathrm{i}\pi(\theta-\frac{k}{2^m})}}\Big| \\ &= \frac{1}{2^{m/2}}\Big|\frac{\sin(2^m\pi\theta)}{\sin((\theta-\frac{k}{2^m})\pi)}\Big|.\end{aligned}$$

The modulus squared coefficients define a discrete probability law with parameter θ. Measurements $(|k\rangle)_{k\in K}$ of the first m-qubits are independent samples of a random variable following the probability law $\frac{1}{2^m}\frac{\sin(2^m\pi\theta)^2}{\sin((\theta-\frac{k}{2^m})\pi)^2}$. Standard

parametric estimations can be used in order to evaluate θ, and hence the eigenvalue. In Figure 8, we plot the result of the *maximum likelihood estimation*. The estimated parameter $\hat{\theta}$ is defined as the maximum of the loss function

$$\mathcal{L}(\theta, K) = \sum_{k \in K} \log \left| \frac{\sin(2^m \pi \theta)}{\sin((\theta - \frac{k}{2^m})\pi)} \right|.$$

This step can be done on a classical computer and does not require to run additional calculations on a quantum computer. The result as well as the estimated variance are given in the following plot for $m = 4$:

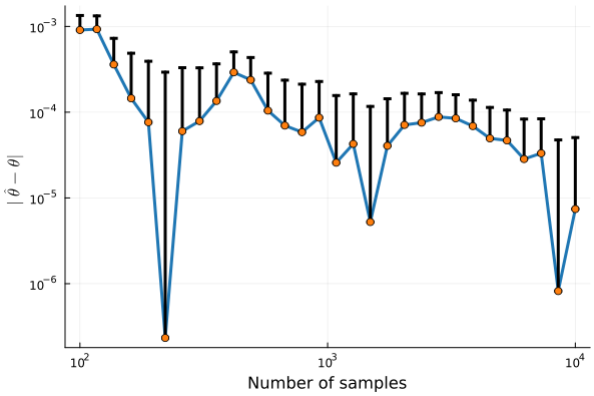


Figure 8: Maximum likelihood estimation. Error bars are determined by computing an empirical variance.

Although the number of qubits to estimate θ is small and a naïve parameter estimation would have an error of order $\frac{1}{2^m}$, the parametric estimation of θ is accurate up to the 4-th digit.

References

[Lin22] Lin Lin. *Lecture Notes on Quantum Algorithms for Scientific Computation*. arXiv:2201.08309 [quant-ph]. 2022. URL: https://arxiv.org/abs/2201.08309.

[NC10] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. 10th Anniversary. Cambridge: Cambridge University Press, 2010.